

Quinn Mair

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EDUCATION

Duke University Pratt School of Engineering

BSE in Electrical and Computer Engineering, Minor in Cultural Anthropology.

Durham, North Carolina

Expected May 2027

- Relevant Coursework: Signal Processing, Digital Systems, Product Design, Entrepreneurship

EXPERIENCE

Automation Systems/EE Design Intern

June 2026 – August 2026

HDR

Denver, CO

- Designed and validated PLC/HMI solutions for industrial water infrastructure, translating client specifications into P&ID drawings, electrical one-lines, control narratives, ladder logic, and operator interfaces
- Programmed and tested PLCs and HMIs using Do-more Designer and Indusoft, validated with a custom PLC I/O rack
- Developed an Excel sizing tool for a major municipality that translates operational bus-fleet metrics into facility footprint requirements
- Collaborated with process, electrical, civil, and project teams to support design reviews for water process and conveyance systems

President

January 2026 – Present

Duke Engineers for International Development

Durham, NC

- Lead planning, budgeting, and stakeholder coordination of 5 infrastructure projects across Bolivia, Eswatini, and the US
- Direct fundraising strategy and outreach that has raised over \$200,000 for international infrastructure projects over 3 years
- Manage a \$70,000+ annual budget, allocating funding across engineering projects and organizational operations

On-Site Construction Manager

May 2025 – August 2025

Engineers In Action

Shiselweni, Eswatini

- Coordinated onsite efforts of 40+ workers through daily meetings and work plans to deliver the longest footbridge in Eswatini (128m)
- Communicated project progress and technical decisions in English and Siswati during construction and inauguration
- Translated designs into excavation drawings using field data and AutoCAD
- First Duke student to be awarded Emerging Leader of the Year for underclassman construction leadership at the annual EIA conference

Co-Founder

May 2024 – September 2025

Lunch Bunch LLC

Durham, NC

- Co-founded a hardware startup developing an interlocking cafeteria tray system to fulfill an LOI from Durham Public Schools
- Drafted and submitted a nonprovisional patent, conducted prior art search, and responded to office actions
- Awarded CFCI GoBig Grant and Duke I&E support grant (x2); featured in Duke Startup Showcase (top 18 of 93)

Teaching Assistant for EGR 105

August 2025 – December 2025

Duke University

Durham, NC

- Taught programming and computational methods to 200+ engineering undergraduates, tailoring lessons to individual students
- Evaluated student work and collaborated with instructional staff to refine course policies

Campus Tour Guide

October 2023 – October 2025

Duke University

Durham, NC

- Delivered tours and presentations to 15-25 prospective students and answered questions across diverse audiences

PROJECTS

32-Bit Pipelined Processor + Video Game

- Designed and verified a 32-bit pipelined processor in Verilog implementing hazard detection, forwarding, carry-lookahead arithmetic, and Modified Booth multiplication
- Debugged processor using GTKWave, Vivado, and Python before FPGA deployment to support a dungeon-crawler video game programmed in MIPS

Pump Station Control System

- Engineered and programmed a PLC/HMI control system for four booster pumps and chlorine injection system, integrating alarms, trends, operator-adjustable setpoints, and real-time equipment status across a 22-screen HMI
- Configured and factory-acceptance-tested the system using a custom PLC/I/O rack, confirming I/O checks and validating alarm and automation behavior

Audio Note Recognition System

- Developed a MATLAB application using Fast Fourier Transform techniques to identify musical notes from complex audio recordings

SKILLS

Controls & Hardware: PLCs, HMIs, Do-more Designer, Indusoft, Logisim, FPGA/Verilog, PCB Design, KiCad

Software & Analysis: MATLAB, Excel, Vivado, AutoCAD, Python